Health-a-thon 2026

FINAL SUBMISSION

Sahara

A palliative-care companion. Doctor-led goals-of-care conversations, written onto the legal form and ready in an emergency.

TEAM NAME Zeros and Ones

TEAM CODE 37MYUP


Shaik Abdul Basith


Shaik Muhammad Awaiz


Dr Sharada Vinod Kutty


Dr Sujay Halkur Shankar


Shirin Ayub

sahara.wedevit.in · Android app and website · English, Hindi, Telugu, Urdu, Kannada

Team Overview & Background



Shaik Abdul Basith

Technical lead

Web, AI, goals-of-care flow



Shaik Muhammad Awaiz

Technical member

Android app, server, testing



Dr Sharada Vinod Kutty

Doctor partner

Pulmonologist · wrote the clinical spec



Dr Sujay Halkur Shankar

Clinician

Pulmonologist, palliative care · set the checklist



Shirin Ayub

Medical student

Symptom tracker

Doctors: chose the problem, review every build

Legal: Rituparna P reviewed the form

Independent team · apps published under WeDevit

Problem Statement and Supporting Evidence



Illustrative scene. A 62-year-old woman with advanced breast cancer.

At the clinic, her doctor discusses her wishes. Only her husband attends.

Two months later, she is unconscious in the Emergency Department. Did she want CPR?

The family disagrees and her earlier note is missing.

15.6 lakh

new cancer cases in India in 2024.

IARC, GLOBOCAN 2024

98.39%

unmet palliative-care need in metastatic cancer.

Patil et al., 2024

88%

of surveyed urban patients wanted to choose their end-of-life treatment.

IndiaSpend, 2019

11.5%

of doctors (45/391) in one hospital were trained in advance directives.

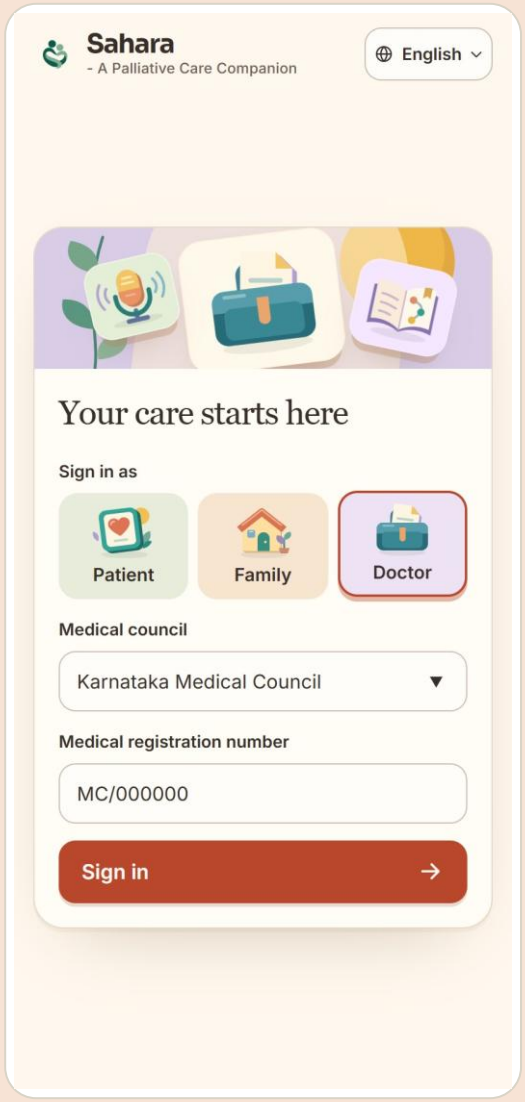
Jacob et al., 2026

What needs to improve

Wishes written on the legal form, signed, and found in seconds in an emergency.

Studies of different groups; not results from Sahara.

Target Users and Use Case



Patient · at home
logs how they feel; wishes written down

Family · at home
same record; reads what the patient said

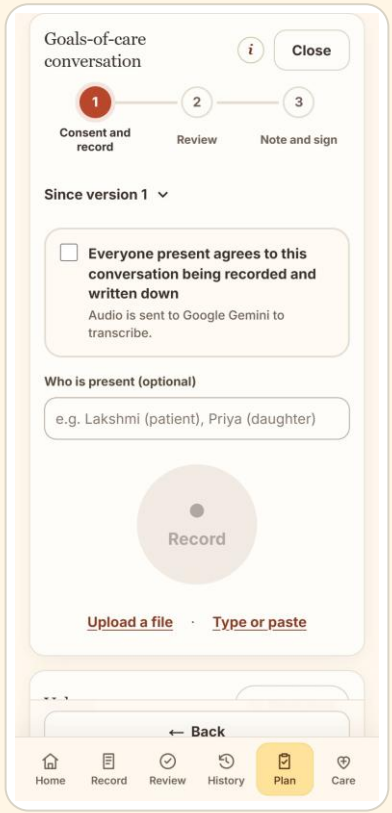
Palliative / oncology doctor · clinic
leads the talk; only reviews and signs

Emergency doctor · ED, minutes to decide
scans the QR card for the wishes

1 app · 3 sign-ins · 5 languages · low-cost Android

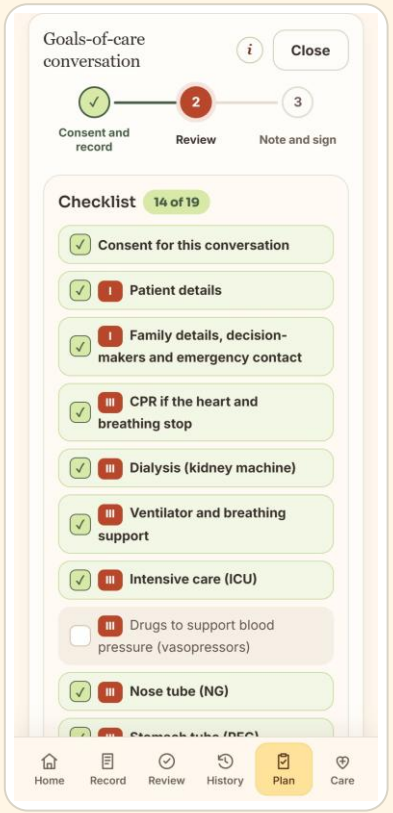
Solution Overview, Key Features & Impact (1/2)

Doctor leads. AI writes it down. The goals-of-care conversation, from consent to a signed note.



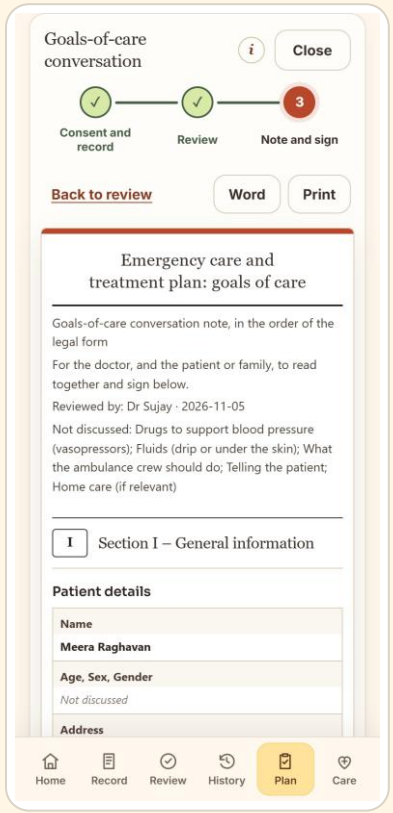
1. Consent, then record

Gemini transcribes (Whisper offline)



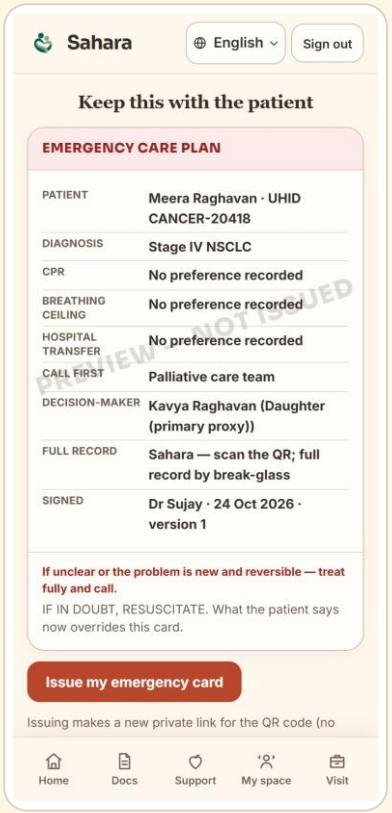
2. Checklist ticks itself

Gaps flagged



3. Written on the legal form

Doctor reviews and edits

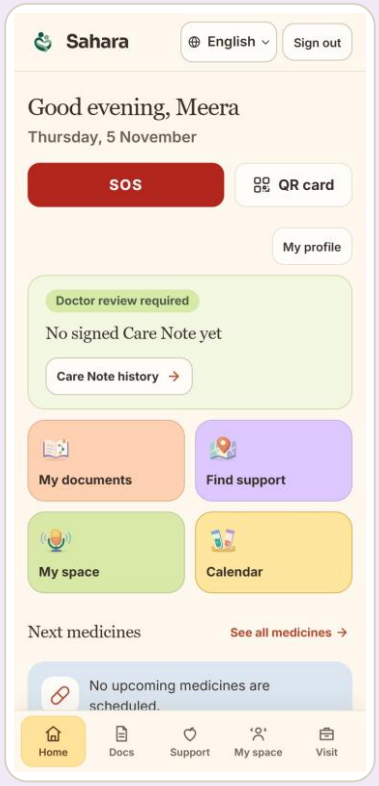


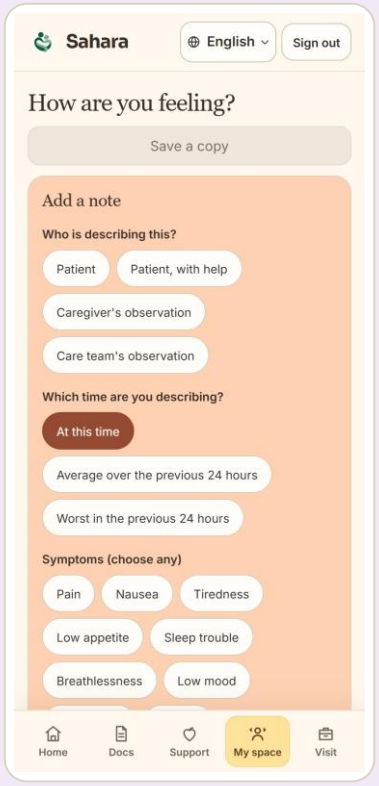
4. Signed, locked, findable

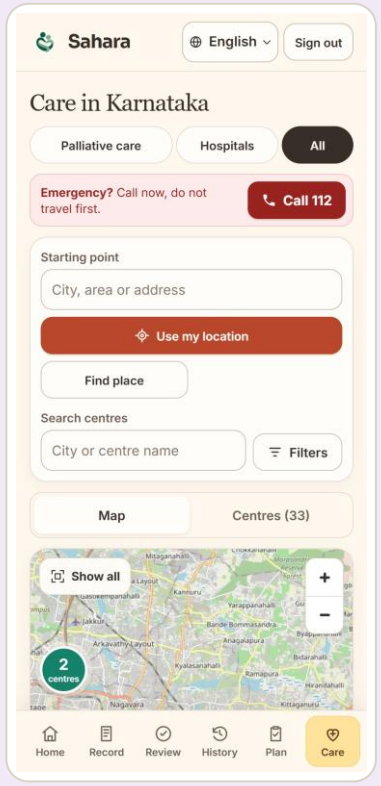
On the emergency QR card

Solution Overview, Key Features & Impact (2/2)

Between visits: a companion for patients and families, shared with their doctor.







How I feel: symptoms, 0–10 distress and a pain body map; trend charts for the doctor.

Medicines: each dose is a tick, with reminders and prescription scan.

Labs and documents: trends, and “Ask your documents” answers with the source line.

Emergency QR card and SOS: wishes, contacts and medicines in one scan.

Help nearby: palliative centres, ambulances and support groups.

Family: invited relatives see and add to the same record.

Expected impact (to be measured in the pilot):

more goals-of-care conversations written and signed · wishes found in seconds in an emergency · fewer family disputes at the bedside · symptoms seen before the next visit.

Technology, Tools and AI Approach



Backup: Gemini busy → next model → Whisper on the phone → typing

AI

- Gemini Flash-Lite / Flash
- OpenAI Whisper (on phone)
- ML Kit translate + read aloud
- On-phone speaker labels

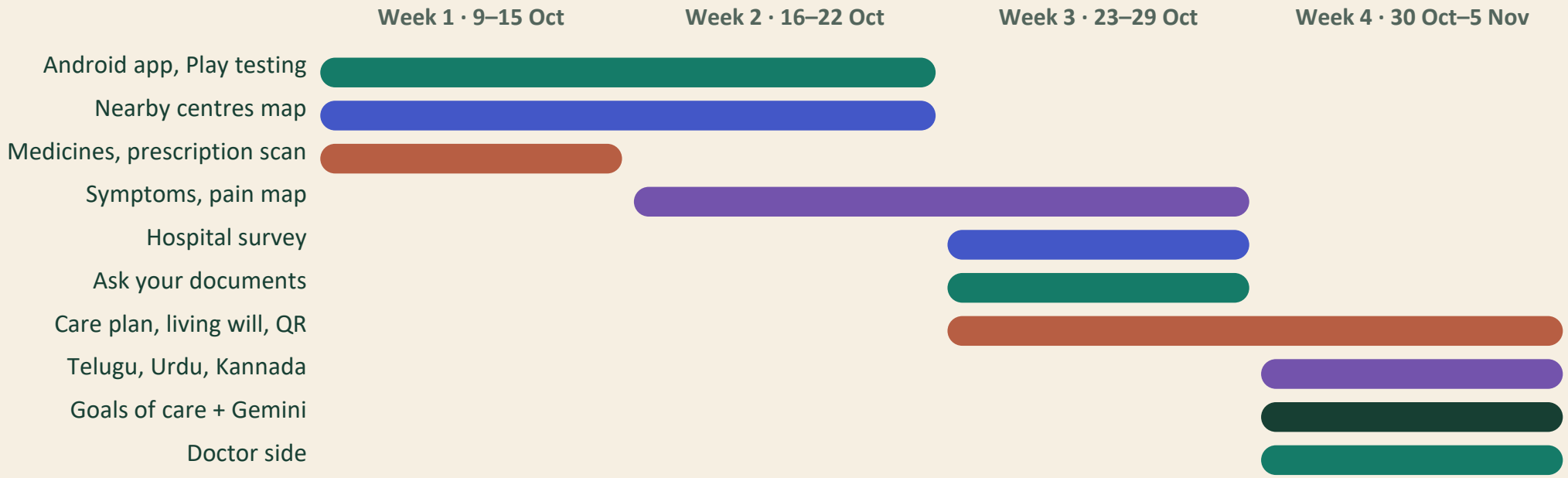
App

- Android · Kotlin, Compose
- Website · Next.js, Vercel
- Supabase · Postgres + RLS
- 5 languages

Data

- Legal goals-of-care form
- Clinical spec (Dr Sharada)
- Pallium India + OpenStreetMap
- FHIR R4 export

Build Sprint Journey and Progress



✓ Working

Patient + family app

Doctor side

Record → signed note

QR card, website live

↻ Evolved

Web prototype → Android

AI note → doctor-led

Legal form, word for word

⚠ Challenges

Gemini “busy” → fallbacks

Legal wording → copied

Faded text → fixed

… Still to do

Real-phone tests

Recording with Dr Sharada

Back-translation

Evaluation and User Validation

877
Android tests pass

594/594
website tests pass

8/8
mock conversations, 6 languages, mapped right

4/4
“Ask your documents” answers correct

Feedback → change

Dr Sujay walkthrough (3 Nov)
tubes & lines, clearer wording, darker text

Legal review (31 Oct)
form copied word for word

Dr Sharada’s spec
12 “always continues” items, word for word

Not yet

real-patient pilot · full real-phone test · speech accuracy on real Indian-language audio · clinician back-translation

Safety and Responsible AI

Privacy and consent

- ✓ Consent tick before recording
- ✓ Gemini only with consent
- ✓ Access by invite · row-level security

AI limits

- ✓ AI never advises treatment
- ✓ Missing topics listed, not guessed
- ✓ Busy model → phone → typing

Human oversight

- ✓ Doctor reviews, edits, signs first
- ✓ Patient or family signs too
- ✓ Locked: any edit voids signatures

Inclusion

- ✓ 5 languages + read aloud
- ✓ Thumb signature, large text
- ✓ Fictional patients in demos

Deployment, Cost and Sustainability Plan

Pilot
1 clinic · our doctors · DPDP consent

Integration
none needed · FHIR export later

Who pays
clinics / NGOs · free for patients

Scale
grants → palliative networks

Item	One-time	Ongoing
Hosting (Supabase + Vercel)	–	₹3,900 / mo
Gemini, per conversation	–	≈ ₹2
Set-up + training	₹25,000	–
Consent + legal review	₹25,000	–
Support	–	₹10,000 / mo
Per clinic (10 clinics)	**₹5,000**	**₹1,600 / mo**

Price: ₹1,500–2,000 per clinic a month

Team estimates. US$1 ≈ ₹86; Supabase Pro $25, Vercel Pro $20 a month; Gemini ≈ $0.30 per million audio tokens, 30 min ≈ 58k tokens; 100 talks per clinic a month.

What Next? Plans to take initiative forward (if any)

1

Prove it

real phones · record with Dr Sharada

2

Pilot 60–90 days

conversations signed · time to find wishes

3

Legal

capacity · partners · withdrawal

4

Fund + grow

grants · palliative networks · FHIR

Share our solution publicly? YES

Slides, brief, video and app description. Not patient records, credentials or private code.

Health-a-thon 2026

Thank you

Zeros and Ones • 37MYUP

workwithdevit@gmail.com | sahara.wedevit.in | Demo video: [add demo video link] | github.com/basithladdu/healthathon (private)



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